

Quiz

March 2, 2026

Which of the following statements regarding the nature and history of operating systems is/are true?

- ☐ A In its earliest iterations, an OS functioned primarily as a library for managing low-level I/O operations.
- ☐ B The Berkeley Software Distribution (BSD) is a derivative of the Linux operating system
- ☐ C A modern OS acts as both a primary resource allocator and a program for controlling the execution of user applications
- ☐ D The term "operating system" typically refers to the kernel as well as the essential system programs that facilitate operation

提交

Which of the following descriptions accurately characterize a computer system?

- ☐ A A computer system includes hardware, operating system, application programs, users
- ☐ B Modern CPUs require instructions to be loaded into main memory (RAM) before execution, as they cannot run instructions directly from a hard drive
- ☐ C In a von Neumann system, the hardware is structured to ensure that executable code and data never occupy the same shared memory space
- ☐ D In the context of computer systems, "users" just mean human beings

提交

Regarding the transition and management of execution modes, which of the following is/are true?

- ☐ A The OS software must provide a mode bit to indicate the current execution environment.
- ☐ B Privileged instructions are restricted to kernel mode; attempting them in user mode results in a trap to the OS
- ☐ C The three primary mechanisms for transitioning from user mode to kernel mode are software-triggered system calls, hardware interrupts, and exceptions
- ☐ D Dual-mode operation is implemented primarily to protect the operating system and other users from potentially malicious or buggy applications

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Identify the true statement(s) regarding the interaction between programs and the kernel:

- A** Interrupts typically occur asynchronously to the current instruction stream, whereas exceptions are synchronous events triggered by specific instructions.
- B** System calls represent a voluntary request for service by a process, while interrupts are signals sent by external hardware devices
- C** Programs invoke system calls using a specific number rather than needing the direct memory address of the kernel function
- D** Although they interface directly with hardware, the specific implementations of traps and interrupts are independent on the underlying hardware architecture

提交

多选题 2分

Which of the following functions are categorized specifically as providing services?

- ☐ A File-system manipulation
- ☐ B Resource allocation
- ☐ C Communications
- ☐ D Accounting

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多选题 2分

Which of the following state transitions are considered valid?

A

Running -> Ready

B

Waiting -> Running

C

New -> Ready

D

Ready -> Waiting

提交

Which statements accurately describe the behavior of the fork() system call?

- ☐ A Following a successful fork(), the parent process receives a return value of 0, while the child receives the PID of itself.
- ☐ B Efficiency is often improved using "copy-on-write" (COW), which delays the actual copying of memory pages until one process attempts to modify them
- ☐ C Through specific system calls, a process in Linux can be assigned to multiple parent processes at the exact same time.
- ☐ D The fork() operation results in the parent's PCB being updated to reflect the addition of a new child

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